

Ultimate Nutrition

Calcium Plus

Just about every cell in the human body needs calcium. Calcium is a mineral that enables blood clotting; aids transmission of nerve impulses, and helps insure a regular heartbeat. The human body relies heavily on calcium to function properly. And, structurally, bones and teeth require calcium to stay strong and healthy... muscle cells too.

Because calcium is so vitally important, our bodies have developed a carefully-regulated process to ensure that new calcium is always -and immediately- available. The body does this in two ways; first, by absorbing calcium directly from the foods and supplements we consume; and, second, when it's not available from digestion, by pulling calcium out of bones.

Bone depletion occurs when blood levels of calcium drop too low, usually when it's been some time since ingesting calcium-containing foods or supplements. And, as the body absorbed calcium away from its bone, the bones become less dense and more fragile.

The main goal of good calcium nutrition is to maintain an adequate intake, so that our bodies do not have to dip into our only calcium reservoir we have – our bones.

Antioxidants work in a variety of ways to reduce the effects of free radicals. They can greatly decrease the damage caused by free radicals, stop them from forming to begin with, or "oxidize" them by combining with them and neutralizing their harmful effects through stabilization. Zinc is an essential nutrient required by humans and animals alike. It plays a variety of biological roles in the body including catalytic, structural, and regulatory roles. Zinc aids in the body's absorption of minerals; including calcium.

Although most people are familiar with calcium's role in bone health, it is important to remember that vitamin D is responsible for helping the body absorb and utilize calcium. The relationship between calcium absorption and vitamin D is similar to that of a door lock and key. Vitamin D is the key that unlocks the door so calcium can leave the intestine and enter the blood stream. Vitamin D also works in the kidneys to help reabsorb calcium that otherwise would be excreted – it makes the whole absorption process more efficient.

That's the plus in Calcium Plus.

Selected References:

Dekkers JC, Van Doornen LJ, Kemper HC. The role of antioxidant vitamins and enzymes in the prevention of exercise-induced muscle damage. *Sports Med* 1996;21(3):213-38.

Krotkiewski M, Gudmundsson M, Backstrom P, Mandroukas K. Zinc and muscle strength and endurance. *Acta Physiol Scand* 1982;116:309–11.

Lukaski HC, Nielsen FH. Dietary magnesium depletion affects metabolic responses during submaximal exercise in postmenopausal women. J Nutr. 2002 May; 132(5): 930-5.

Van Loan MD, Sutherland B, Lowe NM, et al. The effects of zinc depletion on peak force and total work of knee and shoulder extensor and flexor muscles. Int J Sport Nutr 1999;9:125–35.

Frequently asked Questions:

1. What is Zinc? What does it do?

Zinc is not a vitamin. It is an essential trace mineral found in almost every cell in the body. It stimulates the activity of over 100 enzymes that help in the biochemical reactions in the body. It supports a healthy immune system and is needed for wound healing. It also helps to maintain taste and smell, and is needed for DNA synthesis.

2. What are antioxidants?

Antioxidants act as a defense against oxidative damage by helping to inhibit oxidation reactions.